

Name: _____ Date: _____

Collaborators: _____ Lab Section: _____

PRELAB EXERCISES (2 points)*This prelab must be completed and handed in to the lab TA at the start of the lab.***Question 1****1 point**

A 90 kg person jumps from a 30 m tower into a tub of water with a volume of 5 m^3 initially at 20°C . Assuming that all of the work done by the person is converted to heat in the water, what is the final temperature of the water? Note: first find the work done by the person, then find the temperature rise equivalent to that work. Make sure you have the correct value for the mass of the water, and include units.

Question 2**1 point**

When you convert mechanical energy to heat, you are asked to continue taking temperature measurements even after the heat source has been turned off. What effect are we trying to observe, and how do we use this effect in our data analysis?